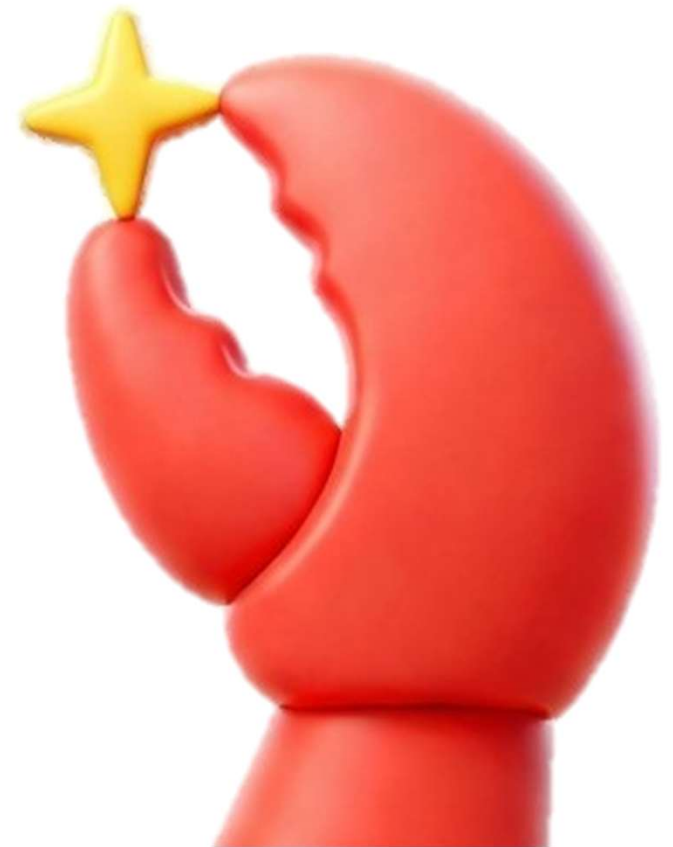


CLASH OF THE **CLAWS**

M S Ramaiah Insititute of Technology – EthosAI **Redefining how AI uses Web**

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Problem Statement

We chose the **GenAI Research** theme because Generative AI is creating a massive global impact across industries, and high-quality data collection forms the backbone of every AI system.

- In today's digital world, many Generative AI companies rely on massive-scale web scraping to train and improve their AI models.
- During this process, website policies such as **robots.txt** and **Terms of Service** are often ignored, leading to unauthorized usage of online content.
- Users increasingly depend on GenAI platforms for research, answers, content generation, and problem-solving instead of visiting original websites directly.
- This shift is significantly reducing website traffic, causing **Publishers, Media Companies, Content Creators, Researchers, and Website Owners** to lose visibility, audience engagement, and advertisement-based revenue.
- The growing misuse of web content is also creating legal, ethical, and sustainability concerns for **AI Companies** themselves.
- To protect their platforms and content, companies such as Cloudflare and Kasada are developing advanced anti-scraping and AI-blocking technologies.
- If this continues, the internet may eventually face a shortage of high-quality publicly accessible data, impacting the future growth of Generative AI itself.

Current Solution & Gaps

Current Solutions

Licensing Deals between AI companies and publishers
(Example: Reddit–Google, OpenAI–News publishers)

Anti-Scraping & AI Blocking by companies like Cloudflare and Kasada

Legal Actions & Lawsuits against AI companies (Example: Reddit vs Anthropic, The New York Times Company vs OpenAI)

Permission-Based Access Models where websites decide crawler permissions

Synthetic Data Generation to reduce dependency on scraped data

Verified AI Crawlers & APIs are being explored for transparent data access

No intelligent system exists that can automatically analyze website permissions, licensing signals, and ethical usage policies at scale. Current approaches are fragmented, inconsistent, and unable to create a sustainable balance between **AI innovation** and **content ownership**.

Existing Gaps

Mostly benefits large publishers; small creators and websites are often excluded

Advanced stealth crawling and disguised bots still bypass protections

Lawsuits are slow, expensive, reactive, and difficult for smaller publishers to fight

No universal adoption or standardized framework yet

Cannot fully replace high-quality real-world human-generated content

Still in early stages with no global standard or enforcement

Our Solution

EthosAI is an AI-driven compliance and decision intelligence system designed to create a sustainable relationship between Generative AI companies and the open web ecosystem.

Instead of blindly scraping data, EthosAI helps AI systems make **intelligent, ethical, and policy-aware decisions** before using website content.

How EthosAI Works

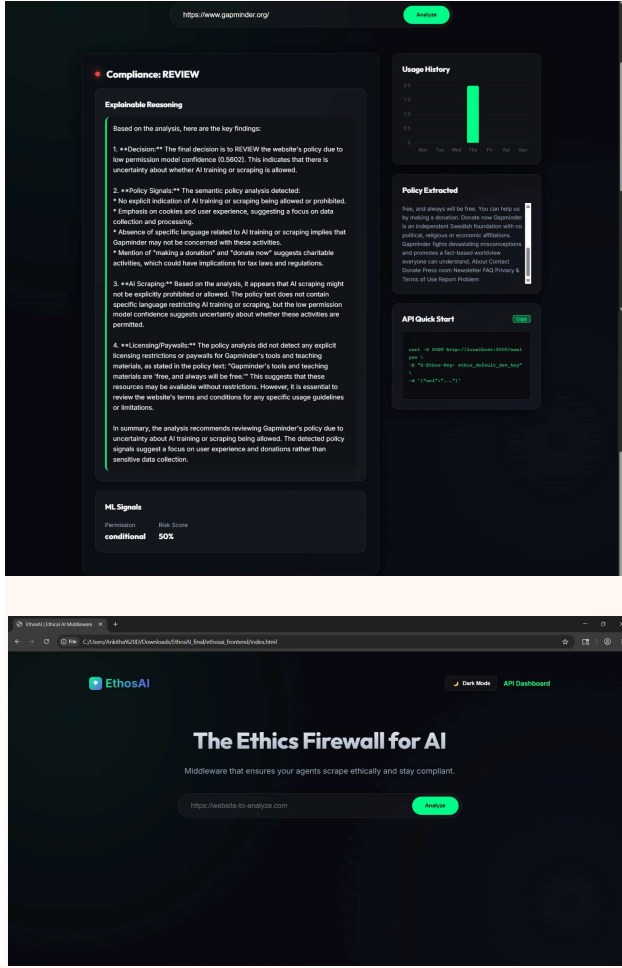
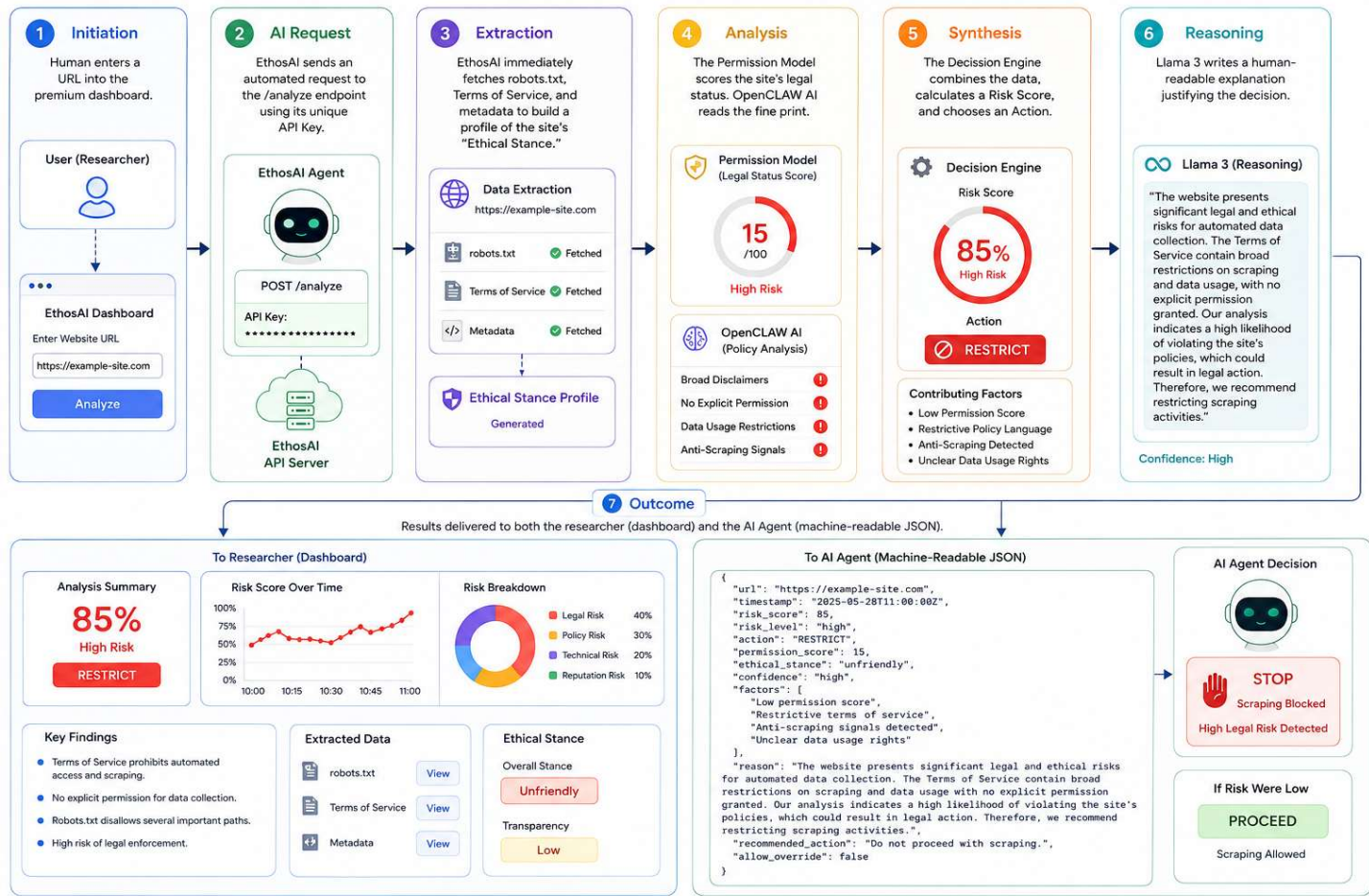
- Analyzes website signals such as: robots.txt, Terms of Service Paywalls, Licensing indicators, Content ownership signals, AI usage permissions
- Uses Machine Learning models to evaluate website accessibility and compliance risk.
- Automatically classifies websites into four decision categories:
 - **Reject** → Do not use content
 - **Cache** → Safe/publicly accessible usage
 - **License** → Requires formal licensing
 - **Negotiate** → Requires publisher agreement or partnership

What Makes EthosAI Different

- Moves AI companies from “**scrape first, face consequences later**” to **intelligent permission-based access**
- Reduces legal, ethical, and compliance risks
- Supports fair compensation and publisher rights
- Helps preserve the long-term sustainability of the open web
- Enables scalable and automated compliance for future AI systems

Vision: EthosAI aims to become the **ethical decision layer for AI data collection**, enabling responsible AI growth while protecting creators, publishers, and the internet ecosystem.

Demo



Tech & Architecture

EthosAI

Tech Stack

Backend (Core): FastAPI (Python), High-concurrency API handling, Automated OpenAPI documentation, Async processing support

Frontend (UI): Vanilla JavaScript, HTML5 & CSS3, Glassmorphism UI design, Chart.js

Machine Learning Layer:

PyTorch — custom multi-class permission classifiers

Hugging Face Transformers — all-MiniLM-L6-v2 for semantic embeddings

Sentence-Transformers — vector similarity search against historical compliance datasets

Reasoning Layer

Ollama

Llama 3 (local deployment)

Human-readable legal reasoning engine

Offline/private inference without external cloud dependency

1. Data Flow (Intelligence Pipeline)

- EthosAI extracts the website domain, robots.txt, and Terms of Service using specialized scrapers.
- The collected text is converted into feature vectors using semantic embeddings and NLP processing.
- Multiple AI models analyze permissions, paywalls, licensing risks, and compliance signals in parallel.
- A similarity engine compares the policy against historical compliance datasets using cosine similarity.
- All results are passed to OpenCLAW for semantic reasoning and conflict detection.
- The system generates a structured compliance and risk analysis report.

2. Model Training

- The compliance model was trained using supervised fine-tuning on legal and policy datasets.
- MiniLM was used as the base Transformer model for lightweight semantic understanding.
- Training data included Terms of Service, robots.txt files, licensing clauses, and restriction patterns.
- The model learns to detect hidden legal signals and ambiguous compliance restrictions.
- A classification layer outputs probability scores for compliance categories.
- Final predictions include Reject, Cache, License, or Negotiate decisions.

3. OpenCLAW (Semantic Policy Analysis)

- OpenCLAW is the legal reasoning engine powered by a locally hosted Llama 3 model.
- It analyzes robots.txt, ToS documents, and licensing clauses semantically.
- The system detects conflicts between scraping permissions and legal restrictions.
- OpenCLAW converts legal language into structured AI-readable compliance logic.
- It provides contextual reasoning instead of simple numerical predictions.
- This enables EthosAI to function as an Ethical Firewall for AI agents.

Impact & Use Cases

Real Life Scenarios:

1. News & Media Protection

The New York Times Company sued OpenAI and Microsoft over unauthorized use of copyrighted articles for AI training.

How EthosAI Helps:

- Detects paywalls and licensing restrictions
- Identifies protected publisher content
- Suggests “License” or “Negotiate” instead of unrestricted scraping

Impact: EthosAI enables ethical and sustainable AI development by helping GenAI companies respect website permissions, licensing policies, and content ownership while collecting training data. It reduces legal and compliance risks, protects publishers and creators through fair usage and licensing, and helps preserve high-quality web data for the future of AI.

2. Social Platforms & Community Data

Reddit filed lawsuits against Anthropic and Perplexity AI for unauthorized scraping and bypassing anti-bot protections.

How EthosAI Helps:

- Respects robots.txt and platform restrictions
- Prevents stealth crawling and unauthorized collection
- Enables permission-aware AI data access

3. Creative & Copyrighted Content

Getty Images sued Stability AI for allegedly scraping millions of copyrighted images without permission.

How EthosAI Helps:

- Detects copyrighted and licensed media
- Flags high-risk datasets before training
- Reduces copyright infringement risks

Scale Potential: EthosAI can scale across GenAI companies, AI search engines, dataset providers, publishers, and enterprise compliance systems. In the future, it can evolve into a standard AI compliance and ethical data governance layer for global AI training pipelines.

Differentiation

Why EthosAI stand out?

1. AI-Native Ethical Decision Engine

Unlike traditional rule-based blockers or simple crawler restrictions, EthosAI uses AI/ML-driven intelligence to analyze website permissions, licensing signals, paywalls, and policy risks at scale. This enables smarter and automated compliance decisions for future AI systems.

2. First-Mover Advantage in AI Compliance Infrastructure

As legal pressure and publisher resistance against unauthorized AI scraping continue to grow, EthosAI positions itself as an early infrastructure layer for ethical AI data governance. Over time, the platform can build a unique compliance dataset and risk-scoring ecosystem that becomes increasingly difficult for competitors to replicate.

3. Network & Data Flywheel Advantage

As more websites, publishers, and AI platforms interact with EthosAI, the system continuously improves its understanding of licensing patterns, compliance risks, and ethical usage behaviors. This creates a powerful data and network effect, making the platform smarter, more accurate, and more valuable over time.

Demo Link

Github Repo Link: <https://github.com/ankithaaa-d/EthosAI.git>

Demo Link: <https://youtu.be/ENCCs561BzI?feature=shared>

Document Submitted (AI Disclosure): <https://drive.google.com/drive/folders/1HFE-x5E5sasiXcwFQQAg-rjyo2LFsRVC?usp=sharing>

Future Implementation

Brownie Slide

API Key Integration (Now)

- The API acts as middleware between AI systems and websites, validating policies before data access is allowed. AI developers can integrate EthosAI into crawlers, search engines, datasets, or autonomous agents with minimal code. FastAPI was used to build high-performance endpoints with automatic OpenAPI/Swagger documentation for easy integration. Responses are returned in structured JSON format containing: decision
- risk_score
- reasoning
- This allows AI systems to make automated compliance decisions in real time. API key authentication ensures secure and controlled access to the platform.
- The API architecture transforms EthosAI into a scalable compliance infrastructure layer for AI systems and autonomous web agents.

Ethos Proxy (Transparent Enforcement Layer) — Future Vision

- Ethos Proxy acts as a reverse proxy through which AI agents route all web requests before accessing websites.
- Every request is intercepted and analyzed in real time for compliance with robots.txt, Terms of Service, licensing clauses, and AI usage policies.
- The proxy can automatically allow, block, restrict, cache, or rate-limit requests based on policy analysis.
- OpenCLAW performs semantic legal reasoning to detect hidden restrictions and conflicts beyond standard robots.txt rules.
- The system prevents unauthorized scraping, commercial misuse, AI training violations, and paywall bypassing.
- AI agents receive structured compliance instructions before extraction begins, enabling safer autonomous browsing.
- Ethos Proxy creates a centralized governance layer with monitoring, logging, audit trails, and policy tracking.
- This transforms EthosAI from a passive compliance tool into an active enforcement and ethical control infrastructure.
- The long-term vision is to build a universal compliance gateway for AI crawlers, search engines, datasets, and autonomous agents.